CHAPTER

Extraction

11

11.1 INTRODUCTION

Literally, the term "extraction" conveys the idea of pulling something out of something else (ex—out, traction—the action of pulling). It is used to indicate a wide variety of actions, from the surgical removal of a tooth to the retrieval of an item from a database. A device for pressing oranges is known as a "juice extractor." In this chapter, however, extraction will be defined as a separation process, based on differences in solubility. A solvent is used to solubilize and separate a solute from other materials with lower solubility in the said solvent.

It is customary to distinguish between two classes of extraction processes:

- *Solid-liquid extraction*, whereby a solute is extracted from a solid phase with the help of a solvent. Examples: extraction of salt from rock using water as solvent, extraction of coffee solubles from roasted and ground coffee in the production of "soluble coffee," extraction of edible oils from oilseeds with organic solvents, extraction of protein from soybeans in the production of isolated soybean protein, etc. Solid-liquid extraction is also termed "leaching" and "elution" (when applied to the removal of adsorbed solute from an adsorbent. See Chapter 12). The mechanism of solid-liquid extraction involves wetting of the solid surface with solvent, penetration of the solvent into the solid, dissolution of the extractables, transport of the solutes from the interior of the solid particles to their surface and dispersion of the solutes within the bulk of the solvent surrounding the solid particles by diffusion and agitation. In some cases, the solubilization step may include chemical changes promoted by the solvent, such as hydrolysis of insoluble biopolymers to produce soluble molecules. The process of extraction with a supercritical fluid (SCE) will be included in this category, it being mainly, but not exclusively, applied to solids.
- *Liquid-liquid extraction*, which is a method for extracting a solute from a solution in a certain solvent, by another solvent. Examples: extraction of penicillin from aqueous fermentation broth by butanol, extraction of oxygenated terpenoids from citrus essential oils using ethanol as a solvent, etc. Liquid-liquid extraction, also known as partitioning, is common in the chemical and pharmaceutical industries and in biotechnology, but much less so in food processing.

Like adsorption (Chapter 12), distillation (Chapter 13), and crystallization (Chapter 14), extraction is a separation process based on molecular transport, in

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which molecules pass from one phase to another under the effect of a difference in chemical potential. The analysis and design of separation operations based on molecular transport require knowledge in three main areas:

1. *Equilibrium:* Net molecular transport stops when equilibrium is reached between the phases, that is, when the chemical potential of the substance in question is the same in all phases in contact with each other. Although true equilibrium can never be reached within a finite duration, the concept of equilibrium is basic to the calculations. In practice, many of such separation processes consist of a number of consecutive stages through which the phases move, following various flow patterns. Countercurrent movement is the most frequent. The common approach is to simulate the real process to a series of theoretical *equilibrium stages* and to correct the deviation from theoretical behavior with the help of empirical or semiempirical *efficiency factors*. Equilibrium data may be given in the form of equations, tables, or charts. Each stage of the extraction process comprises two operations. First, the two phases are brought into contact for a certain length of time, during which mass transfer between the phases occurs. Subsequently, the two phases are separated from each other by a number of possible methods, such as filtration, decantation, centrifugation, squeezing, draining, etc. Physically, the two operations may take place in the same piece of equipment (e.g., an extraction column) or may require two separate devices. The conditions for optimizing the two operations may be mutually contradictory. Thus, division of one of the phases into fine particles improves the rate of interphase mass transfer (see Section 2.7) but makes subsequent separation more difficult.
2. *Material and energy balance*: Each stage of the process must satisfy the laws of conservation of matter and energy, expressed as: *in* = *out* + *accumulation*. In steady-state continuous processes, there is no accumulation. In the graphical representation of the processes, the equations resulting from material balance are represented as *lines of operation*.
3. *Kinetics:* The rate of interphase molecular transport depends on diffusion coefficients and turbulence. Transport kinetics determines the rate at which equilibrium is approached. Kinetic effects are often accounted for with the help of the efficiency factors mentioned above.

11.2 SOLID-LIQUID EXTRACTION (LEACHING)

Solid-liquid extraction is a separation process based on the preferential dissolution of one or more of the components of a solid mixture in a liquid solvent. In this context, the term "solid mixture" is used in its practical meaning. In reality, the physical state of the component to be extracted from the raw material is not always solid. In the case of solvent extraction of oil from oilseeds, for example, the oil is already in the form of liquid droplets in the raw material. In the case of extraction of sugar from beet, the sugar is in solution in the cell juices before contacting the solvent water.

Although the mechanisms involved and the mode of operation are the same in both cases, it is customary to distinguish between "extraction" and "washing." In the case of extraction, the useful product is the extractable solute. In the case of washing, the purpose of the process is to remove undesired solutes from the desired insoluble product. Extraction of coffee solubles in the manufacture of instant coffee is an example of the first case. Removal of lactose and other solutes from cheese curd by washing with water or the removal of low molecular weight components from defatted soybean flour in the production of soybean protein concentrate is an example of the second.

11.2.1 DEFINITIONS

Consider a stage (numbered as Stage n) in a multistage countercurrent solid-liquid extraction process, each stage comprising both the operations of contacting (mixing) and separation. Stage n is represented schematically in Fig. 11.1. Assume that any of the streams of the system consists of some or all of the following three components: the extractable solute (C), the insoluble matrix (B), and the liquid solvent (A).

For the solid streams (slurries), we define:

$E = A + \mathrm{C}$ (kg)
$y = C/(A + C)$ (dimensionless)
$N = B/(A + C)$ (dimensionless)

Note that N is simply the inverse of the solution holding capacity of the insoluble matrix. It depends on the porosity of the matrix and the density, viscosity, and surface tension (wetting power) of the solution, as well as on the method of separation.

For the liquid streams (extracts), we define:

$R = A + C$ (kg)
$x = C/(A + C)$ (dimensionless)
$N = B/(A + C)$ (dimensionless)

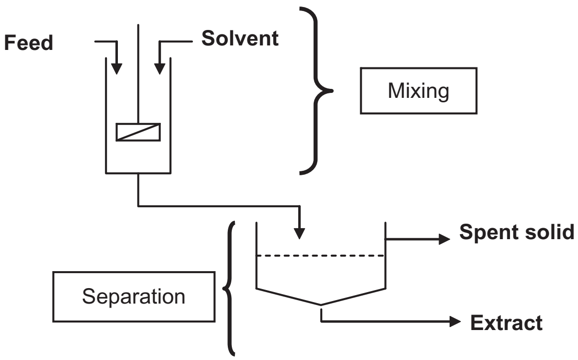

FIG. 11.1

A stage in a solid-liquid extraction process.

Note: In the case of clear extracts (complete solid-liquid separation after mixing), $B = 0$, hence $N = 0$. On the other hand, in turbid extracts, $N > 0$.

The above symbols will be indexed according to the number of the stage from which the represented stream *exits*.

11.2.2 MATERIAL BALANCE

Refer to Fig. 12.1. Material balance for Stage n gives

$$E_{n-1} + R_{n+1} = E_n + R_n \Rightarrow E_{n-1} - R_n = E_n - R_{n+1} \tag{11.1}$$

Since n can be any number, Eq. (11.1) is valid for any of the stages of the system. If the process comprises p stages in total, one can write:

$$E_0 - R_1 = E_1 - R_2 = E_2 - R_3 = \ldots = E_n - R_{n+1} = \ldots = E_p - R_{p+1} = \text{constant} = \Delta \tag{11.2}$$

11.2.3 EQUILIBRIUM

As explained above, it is usually assumed that the streams leaving the stage are in equilibrium with each other. The inaccuracies introduced by this assumption are then corrected using efficiency factors. In the case of solid-liquid extraction, the extract from stage n is assumed to be in equilibrium with the solution imbibed in the slurry leaving the same stage. If the quantity of solvent was sufficient to dissolve all the extractable solute and if the solid matrix is truly inert, then the concentration of the solute in the extract must be equal to that of the imbibed solution. Expressed formally, the equilibrium condition under these assumptions is

$$y_n^* = x_n \tag{11.3}$$

The asterisk indicates that y is the equilibrium concentration.

11.2.4 MULTISTAGE EXTRACTION

The objective of solid-liquid extraction is to extract as much as possible of the solute, using a limited quantity of solvent, so as to obtain a concentrated extract. Obviously, all these conditions cannot be met by single-stage extraction, hence the need for multistage processes. Multistage extraction can be continuous or semicontinuous. A number of options exist with respect to the movement of the solid and the liquid streams with respect to each other (Fig. 11.2). Countercurrent operation, in which the most dilute extract contacts the slurry with the least amount of residual extractables, is the preferred setup.

Eq. (11.2), in conjunction with an equilibrium function such as Eq. 11.3, may be used for the calculation of the composition of the extracts and the slurries from each stage or for the determination of the number of contact stages needed to meet the process specifications. These problems can be solved by means of iterative arithmetic calculations, mathematical modeling (Veloso et al., 2005), or graphically. Graphical methods, despite their lack of accuracy, are extremely useful for the analysis of different separation processes based on multiple contact stages. Later on, we shall see their application in adsorption and distillation. Here, one of the many

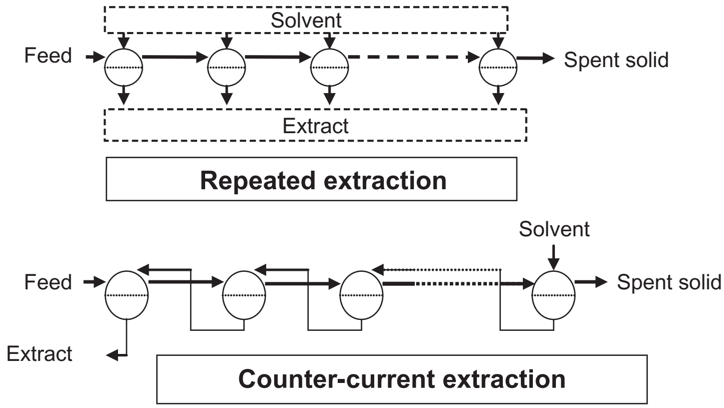

FIG. 11.2

Repeated (cross-flow) and countercurrent extraction processes.

graphical methods for the treatment of countercurrent multistage extraction processes, known as the Ponchon-Savarit method, will be described.

Consider a continuous countercurrent multistage solid-liquid extraction process, consisting of p stages (Fig. 11.3). The raw material to be extracted is fed to Stage 1. Fresh solvent is fed to Stage p. Spent solids are discharged from Stage p. Final extract is collected from Stage 1.

Assume that the process data and specifications are as follows:

1. The relationship $N = f(y)$ is known.
2. Equilibrium is assumed. The equilibrium function $y^* = f(x)$ is known (e.g., $y^* = x$).
3. The extracts are free of suspended solids.
4. The compositions of the starting solid and of the fresh solvent are known.

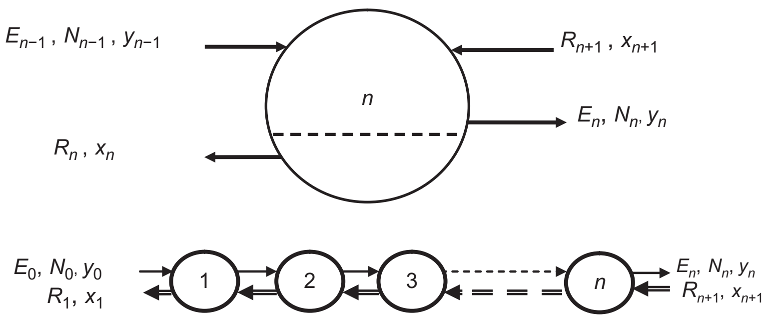

FIG. 11.3

Multistage countercurrent solid-liquid extraction.

5. The feed/solvent mass ratio is specified. The expected yield of recovery for the extractable is also stated. From these data, the composition of the final spent solids and the concentration of the final extract are calculated.

It is desired to calculate the number p of theoretical contact stages needed to achieve the process objectives stated above.

The problem is solved graphically on the Ponchon-Savarit diagram (Treybal, 1980; McCabe et al., 2000), which is a graph of N vs x or y (Fig. 11.4).

- First, the line representing $N = f(y)$ is drawn. This is known as the "slurry line" or the "overflow line."
- The points representing the four streams entering or leaving the system are located. These are the two solid streams $E_0(N_0,y_0)$ and E_p (N_p,y_p) and the two liquid streams R_1 (R_1,x_1) and R_{p+1} (R_{p+1},x_{p+1}).
- Refer to Eq. (11.2). Analytical geometry teaches that the point representing the quantity $E_0 - R_1 = \Delta$ is located on the straight line connecting E_0 with R_1. By the same reasoning, Δ is located on the straight line passing through E_p and R_{p+1}. The point Δ is then marked on the chart, at the intersection of these two straight lines.
- From R_1, the point representing the solid E_1 is found. This point is on the slurry line, with y_1 being related to x_1 by the equilibrium conditions. In our case, it has been assumed that the equilibrium function is as formulated in Eq. (11.3).
- Connecting the point E_1 with Δ, the point R_2 is found on the x, y axis, since $N = 0$ for clear extracts.
- This two-step construction is repeated until the extreme point R_{p+1} is reached or passed. The number of two-step constructions performed is the number of theoretical contact stages.

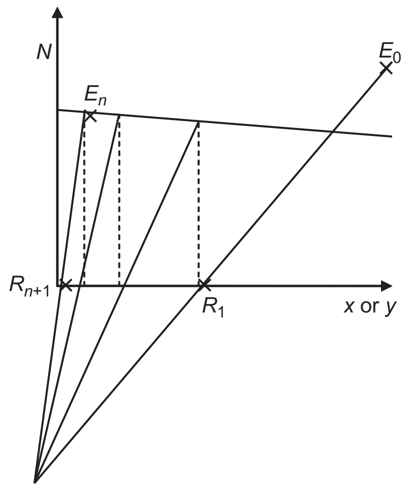

FIG. 11.4

The Ponchon-Savarit graphical representation of multistage extraction.

EXAMPLE 11.1

Biologists have developed a variety of fungus that produces the carotenoid pigment lycopene in commercial quantity. Each gram of dry fungus contains 0.15 g of lycopene. A mixture of hexane and methanol is to be used for extracting the pigment from the fungus. The pigment is very soluble in that mixture. It is desired to recover 90% of the pigment in a countercurrent multistage process. Economic considerations dictate a solvent to feed ratio of 1. Laboratory tests have indicated that each gram of lycopene-free fungus tissue retains 0.6 g of liquid, after draining, regardless of the concentration of lycopene in the extract. The extracts are free of insoluble solids.

What is the minimum number of contact stages required?

Solution

The Ponchon-Savarit diagram is constructed (Fig.11.5) as follows:

- The slurry line is $N = 1/0.6 = 1.667 =$ constant (for all x, y).
- The extract line is $N = 0$ (clear extract)
- Each gram of feed solid has 0.15 g solute, 0.85 g inert matrix, and no solvent. Hence, it is represented by the point: $N = \frac{0.85}{0.15} = 5.67$ and $y = 1$ (point E_0).
- Pure solvent is fed to Stage 1. It is represented by the point: $N = 0$, $x = 0$ (point R_{p+1}).
- Points E_P (spent solids) and R_1 (rich extract) are calculated by a total material balance as follows:

$$\text{In: } 1\text{ kg dry fungus} = 0.15\text{ kg lycopene} + 0.85\text{ kg inert material}$$
$$1\text{ kg pure solvent}$$
$$\text{Out: Spent solids} = 0.85\text{ kg inert mat.} + 0.85 \times 0.6\text{ kg liquid} = 1.360\text{ kg total}$$
$$\text{Rich extract} = 2 - 1.36 = 0.64\text{ kg}$$

Ninety percent of the 0.15 kg of lycopene will be in the extract and 10% in the spent solids. Hence:

$$y_p = \frac{0.15 \times 0.1}{0.85 \times 0.6} = 0.0294 \quad \text{and} \quad x_1 = \frac{0.15 \times 0.9}{0.64} = 0.211$$

- The four points E_0, E_p, R_1, R_{p+1}, are marked on the graph. The difference point Δ is found by joining E_0 with R_1 and E_p with R_{p+1}.
- The equilibrium stages are drawn, assuming $y^* = x$.

  The graph shows between 4 and 5 stages (rounded up to five stages).

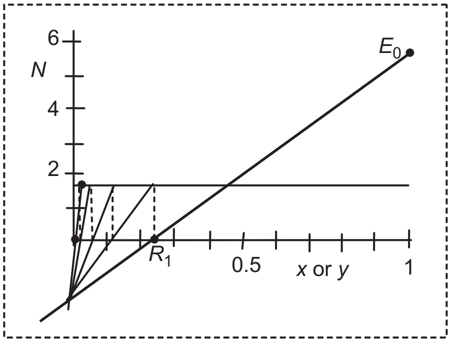

FIG. 11.5

Lycopene extraction example.

11.2.5 STAGE EFFICIENCY

The efficiency of a real contact stage is a measure of the deviation from theoretical (equilibrium) conditions. Stage efficiency in all stage contact processes is generally expressed as "Murphee efficiency," the exact definition of which varies with the process in question (McCabe et al., 2000). In the case of leaching, Murphee efficiency η_M is defined as follows:

$$\eta_M = \frac{X_0 - X}{X_0 - X^*} \tag{11.4}$$

where:

X_0 = concentration of the extract entering the stage
X = actual concentration of the extract leaving the stage
X^* = concentration of the extract that would be reached at equilibrium

The efficiency of a contact stage depends on the time of contact between the phases, the contact surface area, and agitation. Theoretically, it can be calculated based on mass transfer fundamentals. However, exact knowledge of the turbulence conditions, diffusion coefficient, and residence time is usually not available. Stage efficiencies are therefore either determined experimentally or assumed in the light of previous experience. The average Murphee efficiency of a multistage leaching process is also approximately equal to overall efficiency, which is defined as the ratio of the number of theoretical equilibrium stages to the number of actual stages needed to obtain the same end results.

$$\eta_M \approx \eta_{overall} = \frac{N_{theoretical}}{N_{actual}} \tag{11.5}$$

EXAMPLE 11.2

Shelled and dried oil palm kernels contain 47% oil. The kernels are to be extracted with hexane in a countercurrent multistage extractor. The feed to solvent ratio is 1 (i.e., 1 ton of solvent per ton of kernels fed to the extractor). It is assumed that the weight of miscela retained in the kernel is equal to the weight of oil removed. (Miscela is the solution of oil in the solvent). The planned recovery of oil is 99%. The average Murphee efficiency of the contact stages is expected to be 92%. How many contact stages should the extractor have?

Solution

Since the miscela replaces the oil on one-to-one basis, the weight of the spend slurry remains constant and equal to that of the feed.

For all the solids, including the feed, $N = 53/47 = 1.13$
Material balance:

In: 1 kg kernels = 0.47 kg oil + 0.53 kg inert material.
1 kg pure hexane

Out: 1 kg spent kernels, containing 1% of the oil.
1 kg extract, containing 99% of the oil.

The four points are:
E_0: $y = 1$, $N = 1.13$
E_p: $y = 0.01$, $N = 1.13$

R_{p+1}: $x = 0$, $N = 0$
R_1: $x = 0.465$, $N = 0$

The Ponchon-Savarit construction shows five ideal stages (Fig.11.6). The number of real stages will be: 5/0.92 = 5.43. This will be rounded up to six real stages.

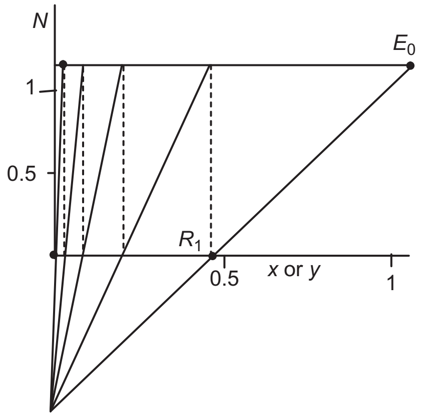

FIG. 11.6

Palm kernels extraction example.

11.2.6 SOLID-LIQUID EXTRACTION SYSTEMS

Most large-scale solid-liquid extraction processes in the food industry are continuous or quasicontinuous. Batch extraction is used in certain cases such as the extraction of pigments from plants, isolation of protein from oilseeds, or the production of meat and yeast extracts. In its simplest form, a batch extraction system consists of an agitated mixing vessel where the solids are mixed with the solvent, followed by a solid-liquid separation device. Decanter centrifuges are advantageously used as separators.

In continuous multistage extraction, the main technical problem is that of moving the solids from one stage to another continuously, particularly if the process is carried out under pressure. The different solid-liquid extraction systems in use differ mainly in the method applied for conveying the solid stream from stage to stage.

1. *Fixed bed extractors*: In fixed bed extractors, the solid bed is stationary. The countercurrent effect is achieved by moving the position where the fresh solvent is introduced. An example of fixed bed extractor is the quasicontinuous, countercurrent, high-pressure process used for the extraction of coffee in the manufacture of instant (soluble) coffee. The system consists of a battery of extraction columns or “percolators” (Fig. 11.7), conventionally six in number.

   Suppose that, at the start of the process, five of the percolators are full with fresh roasted and ground coffee. Percolator No. 6, also full with fresh coffee, is on

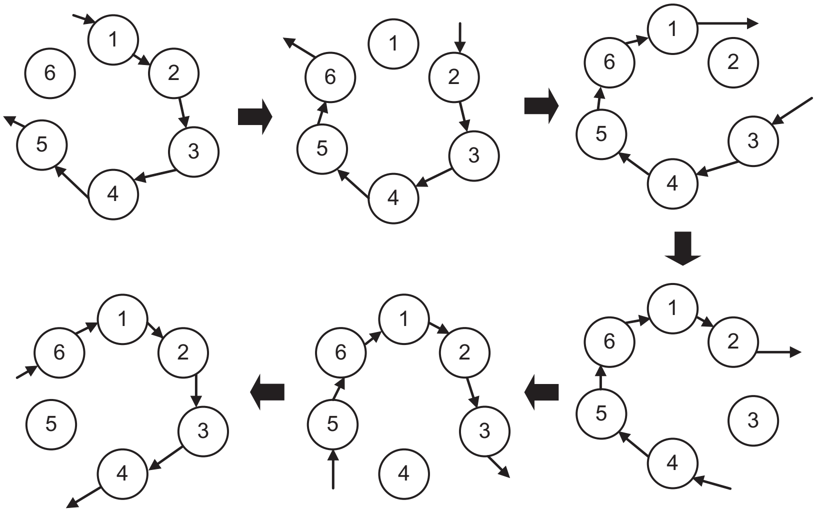

FIG. 11.7

Sequence of stages in semicontinuous countercurrent extraction of coffee.

standby. Very hot water (over 150°C) is introduced into percolator No. 5 and trickles down the bed of coffee, extracting the soluble substances. The entire battery is maintained under high pressure, corresponding to the high temperature of extraction. The extract from percolator No. 5 is pumped to percolator No. 4, to be further enriched in solubles, and so on following the sequence (5-4-3-2-1) through the battery of extractors. The most concentrated extract exits at Percolator No. 1. The coffee in No. 5, having been extracted with pure water, is the one with the lowest concentration of extractables. When its contents are thoroughly exhausted, No. 5 is disconnected, emptied by blowing out the spent coffee solids, cleaned and filled with fresh coffee, while at the same time extractor No. 6, full with fresh coffee, is connected to the tail end of the battery. Hot water is now admitted to No. 4 and the liquid flow sequence becomes 4-3-2-1-6. At this stage, No. 5 becomes the standby unit. At the next stage, No. 4 is emptied, No. 5 is connected and the sequence of liquid flow becomes 3-2-1-6-5, and so on. Thus, the countercurrent effect is achieved without having to move the solids from one stage to another.

2. *Belt extractors*: Belt extractors are extensively used for extracting edible oil from oilseeds and sugar from crushed sugarcane. The material to be extracted is continuously fed by means of a feeding hopper, so as to form a thick mat on a slowly moving perforated belt (Fig. 11.8). The bed height is kept constant by regulating the feed rate. The mat is continuous but distinct extraction stages are delimited by the way in which the liquid stream is introduced. Fresh solvent is sprayed on the solid at the tail "section" of the extractor, that is, at the section nearest to the discharge outlet for spent solids. The first extract is collected at the

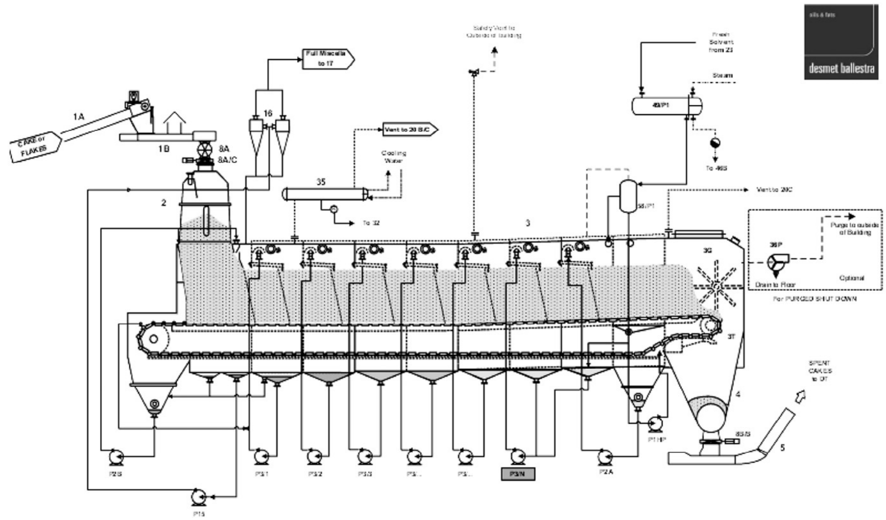

FIG. 11.8

Belt extractor.

Courtesy of Desmet Ballestra A.G.

bottom of that section and pumped over the section preceding the last. This process of spraying the liquid over the solids, percolation of the liquid through the bed thickness, collection of the liquid below the perforated belt, and pumping to the next section is repeated in the direction opposite to the movement of the belt. The most concentrated extract, known as "full micelle," is collected at the bottom of the first (head) section. In their passage from one section to another, the extracts may be reheated with the help of heat exchangers. In the case of extraction with volatile solvents, the entire system is enclosed in an airtight case where a slight negative pressure is maintained to prevent leakage of solvent vapors. In the so-called "two-stage" extractor, two belts in series are used. The transfer of the solids from one belt to the next causes mixing and resettling of the bed. Belt extractors are high capacity units (e.g., 2000–3000 tons per day in the case of solvent extraction of flaked soybeans).

3. *Carrousel extractors*: Carrousel extractors are also most commonly used for the extraction of edible oil from oilseeds (Weber, 1970). These extractors consist of a vertical cylindrical vessel inside which a slowly revolving concentric rotor is installed (Fig. 11.9). The rotor is divided into segments by radial partition walls and rotates over a slotted bottom. The segments contain the solid to be extracted. The liquid extractant is introduced at the top and percolates through the solid bed. The extract exits through the slotted bottom and is collected in chambers separated by weirs, to be pumped back onto the solid bed at the next section. The sequence of liquid collection and pumping is in the direction

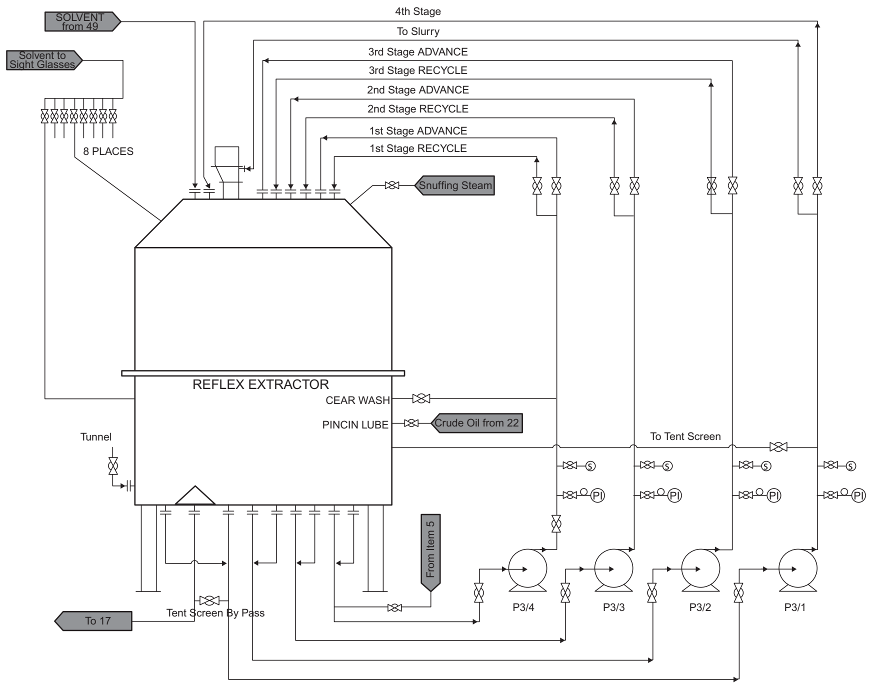

FIG. 11.9

Carousel extractor.

Courtesy of Desmet Ballestra A.G.

opposite to the rotation. At the end of one rotation, the segment passes over a hole in the bottom plate through which the spent solids are discharged. At the next station, the chamber is filled with fresh material and the cycle continues. Carrousel extractors are designed for capacities comparable to that of belt extractors.

4. *Auger extractors*: In auger extractors (Fig. 11.10), the solids are conveyed vertically by a large screw conveyor rotating inside a cylindrical enclosure, against a descending stream of extractant liquid. Inclined versions of this class of extractors are also available. Variants of the auger extractor (often under the name of "diffusers") are extensively used for the extraction of sugar from cut sugar beet chips (cossettes) with hot water.
5. *Basket extractors*: In basket extractors, the material to be extracted is filled into baskets with perforated bottoms. The baskets are moved vertically or horizontally. In the case of vertical basket extractors, also known as bucket elevator extractors, the solvent flows down by gravity through the buckets and is collected at the bottom of the chain. Vertical basket extractors are among the first large-scale continuous solid-liquid extractors (Berk, 1992).

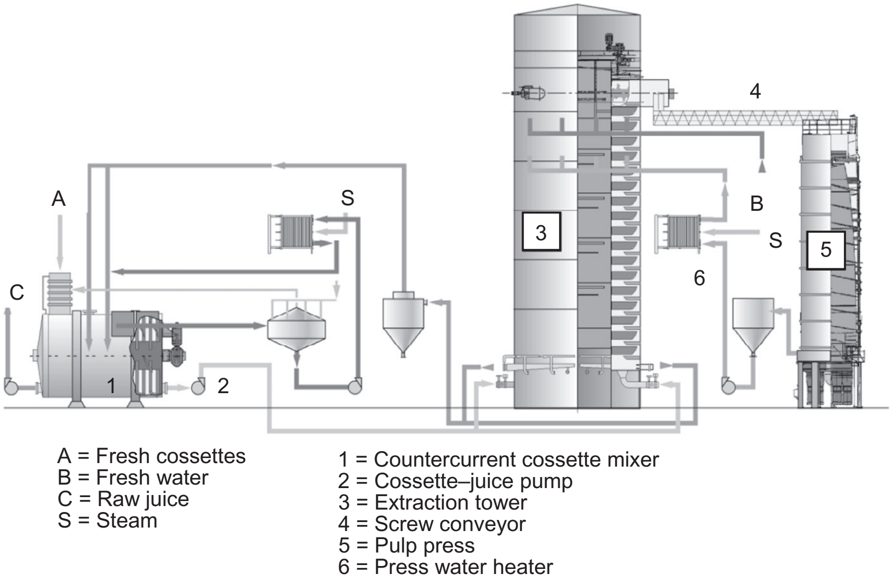

FIG. 11.10

Sugar beet cossette extractor.

Courtesy of BMA Braunschweigische Maschinenbauanstalt A.G.

11.2.7 EFFECT OF PROCESSING CONDITIONS ON EXTRACTION PERFORMANCE

Processing conditions strongly affect the rate and yield of extraction as well as the quality of the extracted product. Understanding the influence of processing parameters on performance is essential for the optimal design and operation of extraction processes and systems.

1. *Temperature*: Where thermal damage is not an issue, high temperatures are preferred for their positive effect on yields and rate. At high temperature, the solubility of the extractables in the solvent is higher and solvent viscosity is lower, resulting in enhanced wetting and penetration capability and higher diffusion coefficients. In the case of volatile, inflammable solvents such as hexane, ethanol, or acetone, safety considerations determine the highest applicable temperature. In certain cases, lower temperatures may be preferred if the selectivity of the solvent toward the desired extractable is improved by lowering the temperature. Ohmic heating may be used for raising the temperature (Pereira et al., 2016).
2. *Pressure*: Solid-liquid extraction at very high temperature implies pressurization to maintain the solvent in liquid state. As mentioned above (Section 2.6), water at high temperature-high pressure is utilized for the extraction of coffee.

The polarity of water decreases as its temperature is raised. Cacace and Mazza (2007) applied pressurized liquid water at very high temperature (pressurized low polarity water) for the extraction of biologically active substances from plant tissue. Corrales et al. (2009) extracted anthocyanin pigments from grape skins, using high (600 MPa) hydrostatic pressure. Other applications of pressurized solvents in extraction processes have been reviewed by Pronyk and Mazza (2009).

3. *Particle size*: The rate of extraction is improved by reducing the size of the solid particles. For this reason, sugar beet is sliced into thin strips (cossettes) and soybeans are ground and flaked prior to extraction. Size reduction facilitates both the internal transport of solute (by reducing the distance to the surface) and the external transport (by increasing the contact area with the solvent). Extraction of proteins from soybean flour is also improved by size reduction (Vishwanathan et al., 2011). In the case of solids of plant proteins, the structure can be disintegrated using cellulolytic and pectolytic enzymes. Flavorings and pigments have been obtained from plant material by enzyme-assisted extraction (Sowdhagya and Chitra, 2010).
4. *Agitation*: Agitation accelerates external transport to and from the particle surface, but does not affect extraction rate if the rate-controlling factor is internal diffusion (Cogan et al., 1967).
5. *Ultrasound-assisted extraction*: Intense sonication of the fluid facilitates release of intercellular materials from suspended solids by disrupting cell walls and promoting turbulence (Yanık, 2017). Ultrasonication is a widely applied laboratory technique for the extraction of enzymes from biomass. Systems for ultrasound-assisted solvent extraction at pilot-plant scale are commercially available.
6. *Applicaton of pulsed electric field (PEF) to extraction*: Pulsed electric fields are known to open pores in cell membranes (electroporation). Originally investigated as a method for the preservation of foods by inactivating microorganisms, PEF treatment has been applied lately as a means to improve solid-liquid extraction yield and extract purity (Loginova et al., 2010, 2011a,b; Yan et al., 2012).
7. *Surfactant-assisted extraction:* Do and Sabatini (2011) extracted oil from peanut and canola, using water as a solvent in the presence of surfactants. Oil recovery yield, in a two-stage semicontinuous process at pilot plant scale, was over 90%.

11.3 SUPERCRITICAL FLUID EXTRACTION

11.3.1 BASIC PRINCIPLES

A supercritical fluid (SCF) is a substance at a temperature and pressure above those of the critical point. The supercritical region for a pure substance (in this case, carbon dioxide) is shown on the phase diagram in Fig. 11.11.

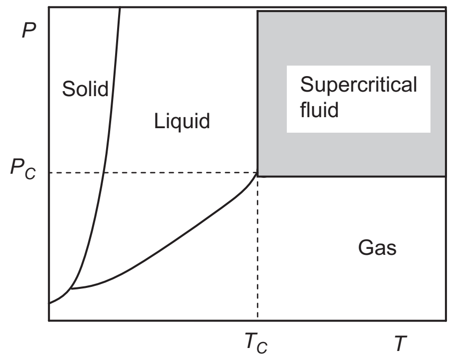

FIG. 11.11

Phase diagram showing supercritical region.

The critical temperature T_C is the temperature above which a gas cannot be liquefied at any pressure. Hence, the critical point C represents the end of the gas-liquid equilibrium curve on the temperature-pressure plane. The density of supercritical fluids is close to that of the liquid while their viscosity is low and comparable to that of a gas. These two properties are key to the functionality of SCFs as extractants. The relatively high density imparts to SCFs good solubilization power while the low viscosity results in particularly rapid permeation of the solvent into the solid matrix.

Supercritical fluid extraction (SCFE or SFE) is an extraction process carried out using a supercritical fluid as a solvent (King, 2000; Brunner, 2005). Although a number of substances could serve as solvents in SFE, carbon dioxide is by far the most commonly used extraction medium. Carbon dioxide near its critical point is a fairly good solvent for low molecular weight nonpolar to slightly polar solutes. However, the solubility of oils in supercritical CO_2 is considerably lower than in conventional hydrocarbon solvents. Nevertheless, in view of its other advantageous characteristics, the use of SFE for the extraction of lipids and fatty acids is not to be dismissed (Sahena et al., 2009; Pradhan et al., 2010; Döker et al., 2010).

Carbon dioxide is nontoxic, nonflammable, and relatively inexpensive. Its critical temperature is 31.1°C (304.1 K), which makes it particularly suitable for use with heat sensitive materials. Its critical pressure is relatively high (7.4 MPa), hence, the high capital cost of the equipment.

11.3.2 SUPERCRITICAL FLUIDS AS SOLVENTS

The solvation capability of SCFs is of considerable economic importance as it determines the extraction yields and the mass ratio of solvent to feed, hence the physical size of the system and the operating cost.

The capability of a supercritical fluid to dissolve a certain substance is approximately represented by a value called "solubility parameter" (Rizvi et al., 1994).

The solubility parameter δ is related to the critical pressure, the density of the gas, and that of the liquid, as follows:

$$\delta = 1.25 P_C^{0.5} \left(\frac{\rho_g}{\rho_l} \right) \tag{11.6}$$

At low pressure, the density of a gas is low while the density of the liquid is almost independent of the pressure. Hence, at low pressure, the solubility parameter is low. The density of the gas increases with the pressure and at the critical point it reaches its maximum value, which is the density of the liquid. Near the critical point, the performance of an SCF as a solvent is then strongly affected by the pressure. The relationship between the solubility parameter and the pressure follows an S-shaped curve (Rizvi et al., 1994), as shown in Fig. 11.12. The pressure-dependence of the SCF is the phenomenon on which supercritical extraction is based. The feed material is extracted with the SCF at a pressure and temperature corresponding to the maximum solubility. The charged solvent (gaseous solution) is separated and the solute is then precipitated from that solution, simply by lowering the pressure.

The effect of temperature on solubility in SCFs is more complex, because the temperature affects both the density of the solvent and the molecular mobility in general. For example, it was found that the solubility of evening primrose oil in supercritical carbon dioxide decreases with temperature at a pressure of 10 MPa, but increases with temperature if the extraction is carried out at 30 MPa (Lee et al., 1994). At intermediate pressure, the solubility is almost independent of the temperature.

The dissolution power of supercritical carbon dioxide can be increased considerably by adding a small amount of another fluid, known as co-solvent, modifier, enhancer, or entrainer. The principal function of the entrainer is often to modify the polarity of SCF CO_2 so as to increase the solubility of more polar substrates. Thus, for example, the solubility of caffeine in supercritical carbon dioxide is greatly enhanced by adding a small amount (e.g., 5%) of ethanol to the solvent (Kopcak and Mohamed, 2005). The extraction of lycopene from tomato skins with supercritical carbon dioxide was facilitated by the addition of water and ethanol of canola oil

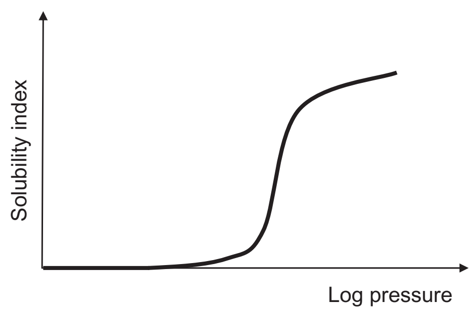

FIG. 11.12

Effect of pressure on the solvation power of SCF.

as modifiers (Shi et al., 2009). The co-solvent, being of lower volatility, remains in the product when the CO_2 is vaporized and can be removed by a subsequent step of desolventizing or by another separation process (Corzzini et al., 2017).

11.3.3 SUPERCRITICAL EXTRACTION SYSTEMS

Most industrial and laboratory scale SCF extractions are batch processes. Owing to the limited solubilization, it is necessary to apply relatively high solvent to feed ratios. In order to supply a high flow rate of fresh solvent, the SCF must be continuously recovered and recycled. The basic components of a carbon dioxide SCE system are (Fig. 11.13):

1. An extraction vessel where a batch of feed material is treated with the SCF solvent.
2. An expansion vessel or evaporator, where the pressure is reduced, causing the solute to precipitate from the extract-laden CO_2.
3. A separator that serves to separate the product from the CO_2 gas.
4. A heat exchanger (condenser) for cooling and condensing the CO_2.
5. A reservoir or tank for storing the liquid CO_2 and adding make-up CO_2 as necessary.
6. A CO_2 pump (compressor) for pressurizing the liquid CO_2 above the critical pressure
7. A heater for bringing the temperature to the desired level (above the critical temperature) before introducing the SCF into the extraction vessel.

The system can be divided into a high-pressure region (from the compressor to the expansion valve) and a low-pressure region (from the expansion valve to the

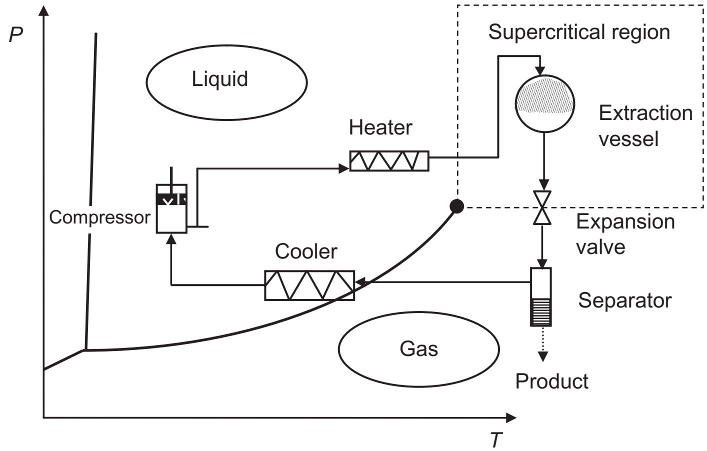

FIG. 11.13

Supercritical extraction system, shown on the CO_2 phase diagram.

compressor). Owing to the high sensitivity of the solubility to pressure change, the pressure drop between the two regions is not very large. The extracted material is often precipitated completely by a slight pressure release. Consequently, the compression ratio and energy consumption of the compressor are usually moderate. In some cases, removal of the extracted material from the charged supercritical CO_2 is carried out by washing the CO_2 with another better solvent (e.g., removal of caffeine from supercritical CO_2 by washing with water).

11.3.4 APPLICATIONS

The advantages of SCF extraction over conventional solvent extraction are:

a. Moderate temperature of operation.
b. Nontoxic, nonflammable, nature-friendly solvents.
c. Very volatile solvents. No solvent residues in the product.
d. Good mass transfer, due to the low viscosity of the solvent.
e. Selective dissolution (e.g., extraction of caffeine from green coffee without removal of flavor precursors).
f. Low overall energy expenditure. No energy consumption for desolventizing.

The disadvantages are:

a. Limited solvation power, requiring high solvent to feed ratio, partially overcome by the use of entrainers.
b. High operation pressure. Expensive equipment.
c. Difficulty to run as a continuous process.

Based on the above, it may be concluded that SCF extraction should be applicable particularly in the following cases:

a. low-to-moderate volume, high value-added processes,
b. heat-sensitive biomaterials, and
c. where selection of a "green technology" is a predominant condition.

The main commercial scale applications are:

a. *Extraction of hops*: Modern breweries make extensive use of hops extracts. Industrial hops extraction is one area where SCF extraction has largely replaced older technologies (Gardner, 1993). SCF extraction has the advantage of extracting mainly the desirable flavor components without the undesirable heavy resins (gums). Furthermore, by varying the extraction conditions, hops extracts richer in aroma or richer in bitter principles can be produced, according to the requirements of the brewing industry. Pelleted hops are extracted with supercritical CO_2 in extraction vessels with capacities ranging from a few hundreds to a few thousands of liters.
b. *Decaffeination of coffee and tea*: This is also an area where SCF extraction has been successful in replacing other extraction methods, mainly because of the absence of

objectionable solvent residues. In the case of coffee, the extraction is carried out on humidified, whole green coffee beans. Both the decaffeinated coffee or tea and the extracted caffeine (after further purification and crystallization) are valuable products. (Lack and Seidlitz, 1993). Decaffeination of green tea by supercritical extraction has also been investigated (Kim et al., 2008).

c. *Other applications*: Other food related applications of SCF extraction include extraction of aromas (Sankar and Manohar, 1994), pigments and physiologically active substances (Shi et al., 2009; Higuera-Ciapara et al., 2005; Rossi et al., 1990; Zeidler et al., 1996), purification of cooking oils (Hong et al., 2010), etc. With the increasing interest in natural nutraceuticals, the use of SCF extraction for the production of substances such as plant antioxidants (Nguyen et al., 1994), phytosterols, omega fatty acids from fish oil (Rubio-Rodriguez et al., 2012), etc. can be expected to grow.

In addition to its industrial applications, SCF extraction is a useful laboratory technique for extraction and isolation. Small-scale complete systems are available for laboratory use.

11.4 LIQUID-LIQUID EXTRACTION

11.4.1 PRINCIPLES

Liquid-liquid extraction, also known as partitioning, is a separation process consisting of the transfer of a solute from one solvent to another, the two solvents being immiscible or partially miscible with each other. Frequently, one of the solvents is water or an aqueous mixture and the other is a nonpolar organic liquid. As in all extraction processes, liquid-liquid extraction comprises a step of mixing (contacting), followed by a step of phase separation. It is important to consider both steps in the selection of solvents and modes of operation. Thus, while vigorous mixing is favorable to the transfer of the extractable from one solvent to the other, it may also impair the ease of phase separation by forming emulsions.

Equilibrium is reached when the chemical potential of the extractable solute is the same in the two phases. Practically, this rule leads to the definition of a "distribution coefficient," K, as follows:

$$K = \frac{C_1}{C_2} \tag{11.7}$$

where C_1 and C_2 are the equilibrium concentrations of the solute in the two phases, respectively. The distribution coefficient is an expression of the relative preference of the solute for the solvents. In ideal solutions (i.e., where the chemical potential may be assumed to be proportional to the concentration), the distribution coefficient at a given temperature is practically constant, that is, independent of the concentration.

In some cases, the efficiency of a liquid-liquid extraction process can be strongly improved by modifying the distribution coefficient. Thus, an organic acid would prefer the nonpolar solvent when not dissociated (i.e., at low pH) and the aqueous solvent when dissociated (i.e., at high pH).

11.4.2 APPLICATIONS

Liquid-liquid extraction is an important separation method in research and chemical analysis. As a commercial process, it is frequently used in the chemical and mining industries and in the downstream recovery of fermentation products (antibiotics, amino acids, steroids). Its applications to food are restricted to isolated cases, such as the transfer of carotenoid pigments from organic solvents to edible oils or the production of "terpeneless" essential citrus oil by extracting the oxygenated compounds of the essential oil with aqueous ethanol.

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